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Unlock a Brighter Future, Transform Your Career with
Accelerated Career Programs in
Embedded Systems Design
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BICARD Faculty



Rushikesh Kulkarni

Experience: 10 Years

Domain Expertise:

C and C++ programming | Firmware Development | Embedded C | Algorithms | Communication Protocols | Bare Metal Programming | Debugging, Testing & Verification



Sridhar Dudum

Experience: 18 Years

Domain Expertise:

Microcontrollers & Microprocessors | Hardware Circuit Design | Assembly Programming Language | Embedded C | Bare Metal Programming | SDLC | FPGA/ ASIC design



Sunil Kumar Singh

Experience: 15 Years

Domain Expertise:

Linux Operating System | Embedded Linux And Device Drivers | Real-Time Operating Systems (RTOS) | Internet of Things (IoT) | Internet Based Technologies



Mohammed Salahuddin

Experience: 20 Years

Domain Expertise:

Hardware Prototyping & Simulation Embedded Software Development | Bare Metal Programming | Debugging, Testing & Verification | Communication Protocols



Ashwini Mate

Experience: 15 Years

Domain Expertise:

Analog & Digital Circuits | Microcontroller & Microprocessors | PCB Design Hardware Testing | Design Patterns for Embedded Systems



Vaibhav Patikar

Experience: 15 Years

Domain Expertise:

Automotive Electronics | CAN FlexRay | Ethernet | LIN On-Board Diagnostics Autosar MATLAB

Faculty at BICARD

innovation meets expertise

Discover BICARD, where innovation meets expertise in education. Our focus is on delivering cutting-edge course content through a passionate and highly competent faculty, ensuring a seamless transfer of knowledge to our students. At BICARD, we recognize the importance of practical, real-time learning that directly impacts our students' careers, empowering them to make a significant difference from day one.

Collaborating on Curriculum Development

Our faculty are not just educators; they are also integral members of BICARD's curriculum development team. Collaborating closely with deans from prestigious academic institutions, they continuously refine our courses to align with the latest research and technological advancements. This dynamic approach ensures that BICARD graduates are equipped with the most relevant skills demanded by today's industries, consistently achieving top scores thanks to the guidance of our experienced faculty.

Domain Experts... All the Way !

Our team of instructors is young, energetic, and deeply passionate about their fields of expertise. Many bring years of hands-on experience from research and consulting projects, enriching their teaching with practical insights gained in the real world. Prior to joining BICARD, faculty members typically boast years of industry experience, further enhancing their ability to bridge academic theory with industry practice.

Excellence Not a Goal But a Standard!

At BICARD, excellence is not a goal but a standard. Our faculty comprises seasoned professionals and subject matter experts, drawn from academia and leading corporations alike. Some faculty members even lead their own Embedded Technology Consulting Businesses, dedicating their valuable insights to our students as visiting lecturers.

Choose BICARD for an educational journey that combines innovation, expertise, and practical wisdom, ensuring you're prepared to thrive in today's dynamic professional landscape.

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Enhance Your Expertise, Safeguard Your Future, Unleash Your Potential, Only with BICARD!

Tailored Industry Training Excellence:

At BICARD, our pledge is to bridge the disparity between candidate skills and industry prerequisites. Our meticulously crafted Post Graduate Diploma Certificate Programs in civil, mechanical, and electronics engineering furnish professionals with the adeptness necessary to excel in today's cutthroat environment.

State-of-the-Art Facilities, Esteemed Educators:

Nestled within our ultramodern 4500 square feet establishment, our programs showcase state-of-the-art training amenities and esteemed instructors hailing from prestigious IITs. With a robust industry linkage, we ensure that your learning journey is both pragmatic and aligned with the latest market demands.

Assured Career Opportunities, Matchless Triumph:

At BICARD, we not only deliver outstanding training but also assure employment, propelling you towards a triumphant career trajectory. Count on us to be your ally in unlocking your full potential and realizing your professional ambitions.

Why Choose BICARD for Information Technology Studies?

We prepare you for Genuine, Hands-On, Industry-Tailored, Job-Ready Positions. Collaborating closely with local businesses and the industry, we ensure our IT and computer engineering programs directly match the skill set required for employment in the IT sector. This guarantees your employability upon program completion. Our industry-endorsed courses are meticulously aligned with the most current industrial trends and standards, acknowledged both locally and internationally.

P. G. Diploma in Embedded Systems Design

Course Overview

Duration: 6 Months

Micro-controller / DSP-based design

- Digital Electronics
- Circuit design and PCB development
- 8 bit/16 bit PIC Microcontrollers
- ARM 7/9/11 Microcontrollers
- Arduino/ESP32
- MSP430/ STM32/Raspberry Pi

Firmware, Driver and Software Development

- Computer Architecture, operating systems
- C programming & Data Structures
- Embedded 'C' with PIC/ARM/stm32
- C++ Programming with ARM CORTEX
- Linux Shell Programming
- Linux porting on ARM Cortex M3/ M4
- Embedded Linux Application on ESP32
- Embedded Python on Arduno / ESP32/STM32
- Linux Device Drivers CHAR/BLOCK
- RTOS on ARM CORTEX –Free RTOS

Communication interfaces

- Communication Protocols
- Wireless technology
- Serial Comm. RS-232, RS-422, RS-485 etc.
- Synchronous Interface: I2C, SPI, SSC and ESSI
- Automotive interface, CAN, LIN, FlexRay
- Memory Devices – Flash based ROM, Storage Cards
- General Purpose Input / Output (GPIO)
- Graphic & Character LCDs
- Real Time Clocks
- Timers: PLL(s), Input Capture / Compare
- Time Processing Units
- Touch Screens, Keypads
- Debugging Interfaces: JTAG, ISP, ICSP
- Universal Serial Bus (USB) Peripherals
- Analog to Digital / Digital to Analog (ADC/DAC) Interfaces ,
- GSM Modules ,Intelligent Power Systems
- Smart Card Readers

Embedded Communication / Linux Network Programming

- Networking Architecture in Linux
- TCP/IP de facto model and layer in kernel
- Client – Server Programming API
- TCP, UDP, RAW, UNIX, FTP,TFTP
- VOIP Protocol Programming
- Iterative server, Concurrent Server
- Broadcast Server Programming
- Wireless Architecture and Programming

Wireless Embedded Systems Design

- Protocol Design and Validation
- Network Embedded Systems
- Bluetooth and IrDA
- Wireless Sensor Networks and ZigBee
- Wireless LAN ,IEEE 802.11
- RFID- GSM and GPRS
- Ubiquitous Computing

Internet of Things (IoT) Development

- Concepts of Internet of Things
- Architecture of Internet of Things
- Technology used in Internet of Things
- Communication Networks,
- Computer Networks
- Digital Signal Processing,
- RFID. Sensor Networks, M2M
- Cloud Computing, Information Security
- Project work using STM32/Raspberry Pi

Automotive Embedded

- Automotive Embedded Architecture and Components
- Architecture of ECU
- AUTOSAR Architecture
- Automotive Protocols (CAN , LIN , FLEXRAY)
- Automotive Use cases
- ADAS
- Misra C Standards

Education Eligibility Requirements for a Postgraduate Diploma Course

Applicants must have a Bachelor's degree in Engineering or Technology (10+2+4 or 10+3+3 years) in IT, Computer Science, Electronics, Telecommunications, Electrical, or Instrumentation. Alternatively, a candidate can have a Master's degree (10+2+3+2 years) in Computer Science, IT, or Electronics. Candidates are required to have obtained a minimum of **65%** marks in their qualifying examination.

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30+ Industry Experts

Trained 7,000+ Students

Placed 7000 Students

450 Companies Leveraging Our Platform

Bicard unlocks your engineering potential with innovative learning frameworks. Our cutting-edge platform combines exponential technology and industry-oriented curricula to prepare you for new-age job roles. Whether you are a beginner or a seasoned engineer, our hands-on interactive programs are effective for all.

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